Assignment 2

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Exercise 1

1a)

To test the relationship between smoking and low birth weight we define a null hypothesis (H0): smoking and low birth weight are independent. We also define an alternative hypothesis (H1): smoking and low birth weight are associated. To test these hypothesis', we apply a contingency table.

```
table_smoke <- table(birth_weight$smoker, birth_weight$lowbwt)
```

```
fisher.test(table_smoke)
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  table_smoke
## p-value = 0.187
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##    0.5262123 277.4965268
## sample estimates:
## odds ratio
##   5.388501
```

A Fisher's Exact Test was used instead of the Chi-square test because some expected counts in the contingency table were below 5, making the Chi-square approximation unreliable. The Fisher's test gives a p-value of 0.187. As our confidence level is a = 0.05, we see that 0.187 > 0.05. This result fails to reject the null hypothesis, which means there is no statistically significant relationship between maternal smoking and low birth weight in this data set. Interestingly, the odds ratio suggest a possible increase in the odds of low birth weight among smokers. However, the wide confidence interval indicates substantial uncertainty, likely due to the small sample size.

To test the relationship between the mother having an age of 35 or more and low birth weight we define a null hypothesis (H0): maternal age is 35 or more and low birth weight are independent. We also define an alternative hypothesis (H1): maternal age is 35 or more and low birth weight are associated.

```r
table_mage35 <- table(birth_weight$mage35, birth_weight$lowbwt)
```

```r
fisher.test(table_mage35)
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  table_mage35
## p-value = 0.4737
## alternative hypothesis: true odds ratio is not equal to 1
## 95 percent confidence interval:
##   0.03498564 33.95657780
## sample estimates:
## odds ratio
##   2.149195
```

The Fisher's Exact Test produced a p-value of 0.4737. Since the given significance level is a = 0.05, we observe that 0.4737 > 0.05. Therefore, we fail to reject the null hypothesis, which means there is no statistically significant relationship evidence of an association between the mother being aged 35 or older and low birth weight in this data set.

1b)

```
## Loading required package: Matrix
```

```
## Loaded glmnet 4.1-10
```

```r
y <- birth_weight$Birthweight
X <- model.matrix(Birthweight ~ ., birth_weight)[, -1]
```

```r
cv_lasso <- cv.glmnet(X, y, alpha = 1)
best_lambda <- cv_lasso$lambda.1se
best_lambda
```

```
## [1] 0.1074244
```

```r
lasso_model <- glmnet(X, y, aplha=1, lambda = best_lambda)
coef(lasso_model)
```

```
## 16 x 1 sparse Matrix of class "dgCMatrix"
##                       s0
## (Intercept) -2.82814008
## ID           .
```

```
## Length       0.03242230
## Headcirc     0.06733298
## Gestation    0.05562886
## smoker      -0.01316677
## mage         .
## mnocig       .
## mheight      .
## mppwt        .
## fage         .
## fedyrs       .
## fnocig       .
## fheight      .
## lowbwt      -0.18180920
## mage35       .
```

```r
full_model <- lm(Birthweight ~ ., data = birth_weight)
summary(full_model)
```

```
##
## Call:
## lm(formula = Birthweight ~ ., data = birth_weight)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.47678 -0.20907 -0.02438  0.20960  0.60017
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -3.830e+00  2.169e+00  -1.766  0.08916 .
## ID          -6.808e-05  1.394e-04  -0.488  0.62931
## Length       5.171e-02  3.888e-02   1.330  0.19503
## Headcirc     8.688e-02  2.997e-02   2.899  0.00751 **
## Gestation    9.189e-02  3.219e-02   2.855  0.00834 **
## smoker      -2.320e-01  1.738e-01  -1.335  0.19353
## mage         1.746e-02  2.904e-02   0.601  0.55293
## mnocig       2.143e-03  7.078e-03   0.303  0.76442
## mheight      1.366e-03  1.473e-02   0.093  0.92680
## mppwt        9.371e-03  1.156e-02   0.811  0.42498
## fage        -1.079e-02  1.977e-02  -0.546  0.58985
## fedyrs      -3.636e-03  3.185e-02  -0.114  0.90998
## fnocig       2.041e-03  4.311e-03   0.473  0.63991
## fheight     -1.543e-02  9.951e-03  -1.550  0.13317
## lowbwt      -1.612e-02  2.558e-01  -0.063  0.95023
## mage35      -5.557e-01  3.669e-01  -1.515  0.14194
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 0.3349 on 26 degrees of freedom
## Multiple R-squared:  0.805,  Adjusted R-squared:  0.6925
## F-statistic: 7.156 on 15 and 26 DF,  p-value: 7.497e-06
```

```r
pred_full <- predict(full_model, newdata = birth_weight)
pred_lasso <- predict(lasso_model, newx = X)

mse_full <- mean((y-pred_full)^2)
mse_lasso <- mean((y-pred_lasso)^2)

mse_full
```

```
## [1] 0.06941883
```

```r
mse_lasso
```

```
## [1] 0.1122155
```

The lasso model selected a subset of predictors by shrinking several coefficients to zero. The performance of the lasso model prediction was compared to a full linear regression model using the mean squared error. The full model achieved a lower MSE of 0.069, compared to the lasso model that achieved a MSE of 0.112. This indicates that the full model has better prediction accuracy on this data set, although the lasso provides a simpler model by selecting only the most relevant predictors.

Exercise 2

2a)

```r
custody <- read.table("custody.txt", header = TRUE)

custody$Indigenous <- as.factor(custody$Indigenous)

summary(custody)
```

```
##       Year      Indigenous   Prisoners         Deaths        Population
##  Min.   :1990   No :6      Min.   : 2041   Min.   : 2.00   Min.   :  168317
##  1st Qu.:1991   Yes:6      1st Qu.: 2368   1st Qu.: 7.75   1st Qu.:  180213
##  Median :1992              Median : 7586   Median :22.00   Median : 6666128
##  Mean   :1992              Mean   : 7934   Mean   :22.42   Mean   : 6875031
##  3rd Qu.:1994              3rd Qu.:13364   3rd Qu.:36.00   3rd Qu.:13538806
##  Max.   :1995              Max.   :14501   Max.   :42.00   Max.   :13995940
```

```r
aggregate(cbind(Deaths, Prisoners) ~ Indigenous, data = custody, sum)
```

```
##   Indigenous Deaths Prisoners
## 1         No    218     80708
## 2        Yes     51     14495
```

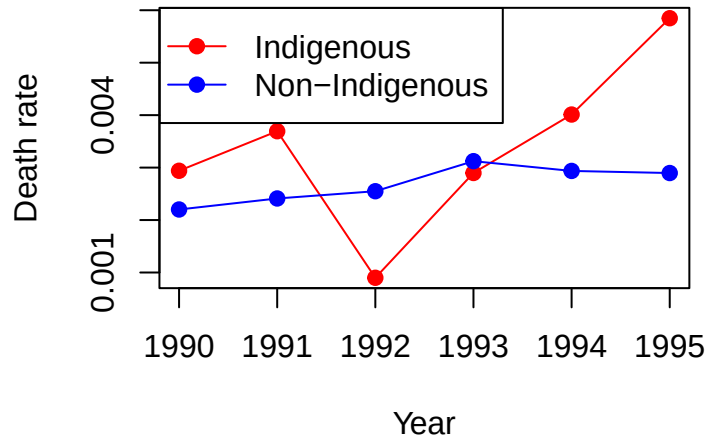
```r
custody$DeathRate <- custody$Deaths / custody$Prisoners
aggregate(DeathRate ~ Year + Indigenous, data = custody, mean)
```

```
##    Year Indigenous    DeathRate
## 1  1990         No 0.0022015656
## 2  1991         No 0.0024115130
## 3  1992         No 0.0025494901
## 4  1993         No 0.0031226766
## 5  1994         No 0.0029366522
## 6  1995         No 0.0028963520
## 7  1990        Yes 0.0029397354
## 8  1991        Yes 0.0036934441
## 9  1992        Yes 0.0008996851
## 10 1993        Yes 0.0028973510
## 11 1994        Yes 0.0040116703
## 12 1995        Yes 0.0058479532
```

```r
aggregate(DeathRate ~ Indigenous, data = custody, mean)
```

```
##   Indigenous   DeathRate
## 1         No 0.002686375
## 2        Yes 0.003381640
```

```r
indig <- subset(custody, Indigenous == "Yes")
nonindig <- subset(custody, Indigenous == "No")

plot(indig$Year, indig$DeathRate,
     type = "o", pch = 19, col = "red",
     ylim = range(custody$DeathRate),
     xlab = "Year", ylab = "Death rate")

lines(nonindig$Year, nonindig$DeathRate,
      type = "o", pch = 19, col = "blue")

legend("topleft",
       legend = c("Indigenous", "Non-Indigenous"),
       col = c("red", "blue"),
       pch = 19,
       lty = 1)
```



The number of prisoners increases slightly from 1990 to 1995 for both Indigenous and non-Indigenous groups, although the Indigenous prison population is much smaller. The number of deaths fluctuates across years, with higher raw counts for non-Indigenous prisoners due to their much larger prison population. A more informative measure is the death rate (Deaths/Prisoners). The exploratory analysis suggests that the death rate is generally higher for Indigenous prisoners, although it varies considerably from year to year. These observations motivate a formal statistical analysis using logistic regression to determine whether the difference between the groups is statistically significant.

2b)

```r
model1 <- glm(cbind(Deaths, Prisoners - Deaths) ~ Year + Indigenous,
              data = custody,
              family = binomial)

summary(model1)
```

```
##
## Call:
## glm(formula = cbind(Deaths, Prisoners - Deaths) ~ Year + Indigenous,
##     family = binomial, data = custody)
##
## Coefficients:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -161.64505   72.17485  -2.240   0.0251 *
## Year             0.07815    0.03622   2.158   0.0309 *
## IndigenousYes    0.25588    0.15587   1.642   0.1007
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 16.4565  on 11  degrees of freedom
## Residual deviance:  9.0151  on  9  degrees of freedom
## AIC: 70.452
##
## Number of Fisher Scoring iterations: 4
```

```r
model2 <- glm(cbind(Deaths, Prisoners - Deaths) ~ Year * Indigenous,
              data = custody,
              family = binomial)

anova(model1, model2, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: cbind(Deaths, Prisoners - Deaths) ~ Year + Indigenous
## Model 2: cbind(Deaths, Prisoners - Deaths) ~ Year * Indigenous
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         9     9.0151
## 2         8     7.7556  1   1.2596   0.2617
```

A logistic regression model was fitted with the number of deaths among prisoners as the response variable, with Year as a continuous predictor and Indigenous status as a factor predictor. An additional model including the interaction between Year and Indigenous was fitted to test whether the time trend differs between the groups. A likelihood ratio test comparing the models gave a p-value of 0.2617. Since this is greater than 0.05, the interaction is not statistically significant. Therefore, there is no evidence that the effect of Year differs between Indigenous and non-Indigenous prisoners.

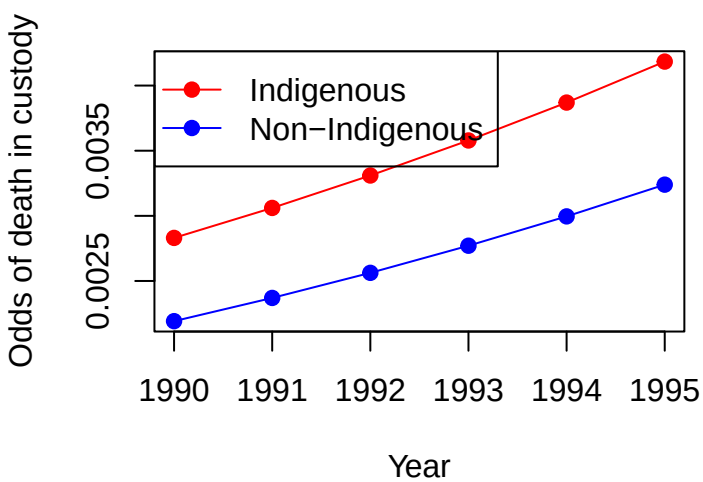
2c)

```r
exp(coef(model1)["IndigenousYes"])
```

```
## IndigenousYes
##      1.291598
```

The estimated odds ratio is 1.29, indicating that the odds of death in custody for Indigenous prisoners are approximately 29% higher than for non-Indigenous prisoners, holding Year constant. Since the interaction between Year and Indigenous was not statistically significant, the odds ratio does not depend on year. Thus, the relative difference between the two groups remains constant over the observed time period.

```r
years <- 1990:1995

newdata_yes <- data.frame(Year = years, Indigenous = "Yes")
newdata_no  <- data.frame(Year = years, Indigenous = "No")

p_yes <- predict(model1, newdata_yes, type = "response")
p_no  <- predict(model1, newdata_no, type = "response")

odds_yes <- p_yes/(1-p_yes)
odds_no  <- p_no/(1-p_no)

plot(years, odds_yes, type="o", col="red", pch=19,
     ylim=range(c(odds_yes, odds_no)),
     xlab="Year", ylab="Odds of death in custody")

lines(years, odds_no, type="o", col="blue", pch=19)

legend("topleft",
       legend=c("Indigenous","Non-Indigenous"),
       col=c("red","blue"),
       pch=19,
       lty=1)
```



The predicted odds from 1990 to 1995 were plotted for both groups, showing higher odds for Indigenous prisoners across all years. The difference between the groups remains approximately constant over time.

2d)

```r
model_pop <- glm(cbind(Deaths, Prisoners - Deaths) ~ Year + Indigenous + Population,
                 data = custody,
                 family = binomial)

anova(model1, model_pop, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: cbind(Deaths, Prisoners - Deaths) ~ Year + Indigenous
## Model 2: cbind(Deaths, Prisoners - Deaths) ~ Year + Indigenous + Population
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         9     9.0151
## 2         8     7.7054  1   1.3098   0.2524
```

```r
model_pop_int <- glm(cbind(Deaths, Prisoners - Deaths) ~ Year + Indigenous + Population + Indig
                     data = custody,
                     family = binomial)

anova(model_pop, model_pop_int, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: cbind(Deaths, Prisoners - Deaths) ~ Year + Indigenous + Population
## Model 2: cbind(Deaths, Prisoners - Deaths) ~ Year + Indigenous + Population +
##     Indigenous:Population
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         8     7.7054
## 2         7     7.6892  1 0.016175   0.8988
```

The model from part (b) was extended by including Population as an additional predictor. A likelihood ratio test comparing the models with and without Population gave a p-value of 0.2524, indicating that Population does not significantly improve the model. An additional model including the interaction between Indigenous and Population was also considered, but this interaction was not statistically significant ($p = 0.8988$). Since neither Population nor the interaction term improved the model, their inclusion is not justified. Therefore, the final model includes only Year and Indigenous as predictors, which is the simplest model not contradicted by the data.

2e)

```r
model_prison <- glm(cbind(Prisoners, Population - Prisoners) ~ Year + Indigenous,
                    data = custody,
```

```r
                    family = binomial)

model_prison_int <- glm(cbind(Prisoners, Population - Prisoners) ~ Year * Indigenous,
                        data = custody,
                        family = binomial)

anova(model_prison, model_prison_int, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: cbind(Prisoners, Population - Prisoners) ~ Year + Indigenous
## Model 2: cbind(Prisoners, Population - Prisoners) ~ Year * Indigenous
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1         9     46.830
## 2         8     16.873  1   29.957 4.418e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

To investigate whether Year and Indigenous status affect the proportion of prisoners relative to the population, a logistic regression model was fitted with the number of prisoners among the population as the response variable and Year and Indigenous as predictors. A second model including the interaction between Year and Indigenous was also fitted. A likelihood ratio test comparing the models gave a p-value of 4.418e-08, indicating that the interaction term is highly significant. This suggests that the change over time in the proportion of prisoners differs between Indigenous and non-Indigenous populations. Therefore, the interaction between Year and Indigenous should be included in the model.

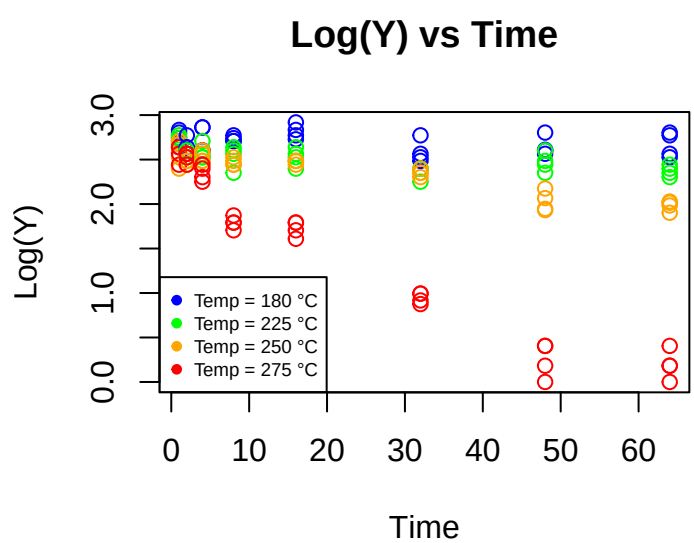
2f)

The exploratory analysis suggested that the death rate among prisoners is generally higher for Indigenous prisoners than for non-Indigenous prisoners, although the rates vary between years. Logistic regression modelling showed that the odds of death in custody for Indigenous prisoners are approximately 1.29 times the odds for non-Indigenous prisoners, holding Year constant. The interaction between Year and Indigenous was not statistically significant, indicating that the relative difference between the groups remains approximately constant over time. Including Population in the model did not significantly improve the model fit, so the simpler model with Year and Indigenous was retained. However, modelling the proportion of prisoners relative to the population showed a significant interaction between Year and Indigenous, indicating that the change over time in imprisonment rates differs between the two groups. Overall, the results suggest that Indigenous prisoners have somewhat higher odds of death in custody, while a much larger disparity exists in the proportion of Indigenous people imprisoned relative to their population size.

Exercise 3

3a)

```
colors <- c("180" = "blue", "225" = "green", "250" = "orange", "275" = "red")
plot(nonlin_regr_nelson$x1, log(nonlin_regr_nelson$y),
     col  = colors[as.character(nonlin_regr_nelson$x2)],
     xlab =  "Time", ylab = "Log(Y)",
     main = "Log(Y) vs Time"
     )

legend("bottomleft",
       legend = paste("Temp =", c(180, 225, 250, 275), "°C"),
       col    = c("blue","green","orange","red"),
       pch    = 19,
       cex    = 0.6)
```



The above scatter plot shows the effect of time and temperature on the dialectric breakdown strength. It is clear from the plot that breakdown strength decreases with time and that there is a strong relation between temperature and the effect of time on the breakdown strength, with lower temperatures remaining similar across time, while higher temperatures, particularly around 275 degrees, drop more drastically over time. This suggests that degradation accelerates with temperature increase.

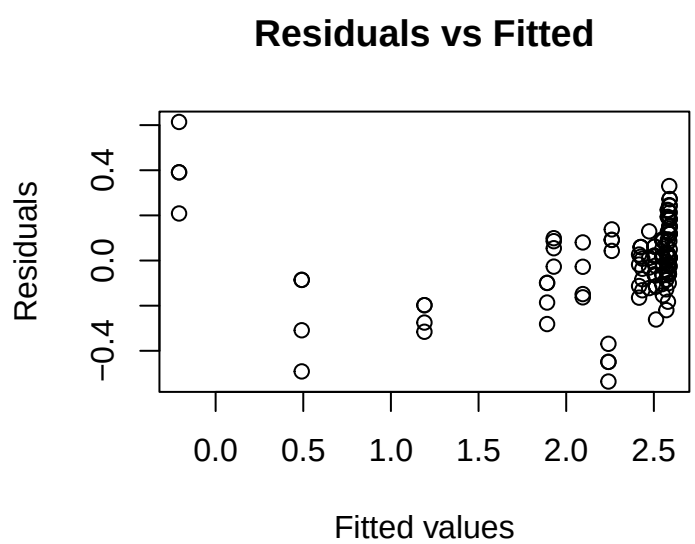
```
fit <- nls(log(y) ~ theta1 - theta2 * x1 * exp(-theta3 * x2),
           data  = nonlin_regr_nelson,
           start = list(theta1 = 2.9, theta2 = 0.001, theta3 = -0.01),
           control = nls.control(maxiter = 1000))

summary(fit)
```

```
##
## Formula: log(y) ~ theta1 - theta2 * x1 * exp(-theta3 * x2)
##
## Parameters:
##          Estimate Std. Error t value Pr(>|t|)
## theta1  2.591e+00  1.915e-02 135.284   <2e-16 ***
## theta2  5.618e-09  6.112e-09   0.919     0.36
## theta3 -5.770e-02  3.957e-03 -14.581   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1743 on 125 degrees of freedom
##
## Number of iterations to convergence: 52
## Achieved convergence tolerance: 5.591e-06
```

```r
RSS <- sum(residuals(fit)^2)
SS_total <- sum((log(nonlin_regr_nelson$y) - mean(log(nonlin_regr_nelson$y)))^2)
R2 <- 1 - RSS/SS_total
cat("R² =", R2)
```

```
## R² = 0.9302059
```

```r
plot(fitted(fit), residuals(fit),
     xlab="Fitted values", ylab="Residuals",
     main="Residuals vs Fitted")
```



The proposed initial values are as follows: theta1 = 2.9, theta2 = 0.001, theta3 = -0.01 The NLS estimates are: theta1 = 2.591, theta2 = 5.618 * 10^-9, theta3 = -0.058 with a residual standard error of 0.174 meaning an estimated

variance of (0.174)^2 = 0.030. The regression model is able to capture the relationship between the breakdown strength degradation, temperature and time. With $R^2 = 0.93$, the model is able to explain 93% of variation indicating strong predictive power and overall a strong model fit. The residuals vs fitted plot shows residuals scattered around zero, indicating that the model captures the overall structure of the data reasonably well. Some clustering is observed at higher fitted values, which likely reflects multiple observations having the same combinations of time and temperature.

3b)

```r
# From Variance-covariance matrix, get the variance and convert to SE
vcov_mat <- vcov(fit)
se <- sqrt(diag(vcov_mat))
est_params <- coef(fit)

z  <- (est_params["theta1"] - 2.5) / se["theta1"]
p <- 2 * pnorm(-abs(z))

cat("Z =", z, "\n")
```

```
## Z = 4.73542
```

```r
cat("p-value     =", p, "\n")
```

```
## p-value     = 2.186018e-06
```

```r
res <- data.frame(
  Estimate = est_params,
  SE       = se,
  Lower    = est_params - 1.96 * se,
  Upper    = est_params + 1.96 * se
)

print(signif(res, 3))
```

```
##         Estimate       SE     Lower     Upper
## theta1  2.59e+00 1.91e-02  2.55e+00  2.63e+00
## theta2  5.62e-09 6.11e-09 -6.36e-09  1.76e-08
## theta3 -5.77e-02 3.96e-03 -6.55e-02 -4.99e-02
```

With a p value less than 0.05, we can reject the null hypothesis as there is significant evidence to suggest that theta1 is not equal to 2.5.The 95% confidence interval range is [2.55, 2.63], which excludes 2.5.

3c

```r
x1 <- 57
x2 <- 260

pred_log_y <- est_params["theta1"] - est_params["theta2"] * x1 *
         exp(-est_params["theta3"] * x2)

grad <- c(
  1,
  -x1 * exp(-est_params["theta3"] * x2),
  est_params["theta2"] * x1 * x2 * exp(-est_params["theta3"] * x2)
)

# Variance and SE using delta method formula
var_f <- t(grad) %*% vcov_mat %*% grad
se_f  <- sqrt(var_f)

lower <- pred_log_y - 1.96 * se_f
upper <- pred_log_y + 1.96 * se_f

cat("Predicted log(y) =", round(pred_log_y,3), "\n")
```

```
## Predicted log(y) = 1.542
```

```r
cat("SE =", round(se_f,3), "\n")
```

```
## SE = 0.058
```

```r
cat("95% Confidence Interval: [", round(lower,3), ",", round(upper,3), "]\n")
```

```
## 95% Confidence Interval: [ 1.427 , 1.656 ]
```

By means of the delta method formula, the estimated mean of log(y) at x = (57,260) is 1.542. The corresponding 95% confidence interval is [1.43, 1.66].

3d

```r
fit_reduced <- nls(log(y) ~ 2.5 - theta2 * x1 * exp(-theta3 * x2),
                   data  = nonlin_regr_nelson,
                   start = list(theta2 = est_params["theta2"],
                                theta3 = est_params["theta3"]),
                   control = nls.control(maxiter = 1000))

summary(fit_reduced)
```

```
##
## Formula: log(y) ~ 2.5 - theta2 * x1 * exp(-theta3 * x2)
##
## Parameters:
##          Estimate Std. Error t value Pr(>|t|)
## theta2  3.898e-10  5.609e-10   0.695    0.488
## theta3 -6.723e-02  5.248e-03 -12.811   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1887 on 126 degrees of freedom
##
## Number of iterations to convergence: 39
## Achieved convergence tolerance: 4.894e-07
```

```r
rss_full    <- sum(residuals(fit)^2)
rss_reduced <- sum(residuals(fit_reduced)^2)

anova_result <- anova(fit_reduced, fit)
print(anova_result)
```

```
## Analysis of Variance Table
##
## Model 1: log(y) ~ 2.5 - theta2 * x1 * exp(-theta3 * x2)
## Model 2: log(y) ~ theta1 - theta2 * x1 * exp(-theta3 * x2)
##   Res.Df Res.Sum Sq Df  Sum Sq F value    Pr(>F)
## 1    126     4.4862
## 2    125     3.7977  1 0.68852  22.663 5.247e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

In order to investigate whether the reduced model with theta1 = 2.5 is appropriate, a comparison is carried out between the full and reduced model using anova test. As the residual sum of squares for the reduced model (4.486) is larger than that of the full model (3.798), it can be concluded that the reduced model fits the data less closely than the full model. Furthermore, the F-value is 22.663 with p-value = 5.25*10^-6 which is less than 0.05 so we can reject the null hypothesis and conclude that the the reduced model is not appropriate as it worsens the model fit.